Axiomatic Derivation of the Leading Valence Engagement Amplitude in the PDL Framework

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Abstract

The Projective Dynamic Logo (PDL) framework derives physical constants from the proton integer quintuplet $(24, 28, 930, 10087, 11017)$ without free parameters. Paper D28 of the PDL corpus established that the proton-to-electron mass ratio correction factorises as $\delta\mu = (155/11017) \cdot (1 - 2\Delta m_{\text{iso}}/m_p) + O(47 \text{ ppm})$, identifying $155/11017 \in \mathbb{Q}$ as the leading PDL valence engagement amplitude, and left its derivation from first principles as open problem OP2 [10]. The present paper resolves OP2 entirely.

We prove two lemmas. *Lemma 1* establishes that a mixed triangle between K_4 (the electron prototype) and a proton vertex is stable across both half-cycles of the K_4 pulsation if and only if the product of cross-edge signs satisfies two conditions: initial alignment with K_4 (condition A) and exact phase-locking with the pulsation (condition B). The proof is a two-line algebraic substitution, verified exhaustively over all coherent K_4 configurations. *Lemma 2* establishes that $r_{\text{val}} = 2r_u + r_d = 930$ is the unique integer engagement of valence cores satisfying both net charge $+1$ and divisibility by $\Re = 6$, the latter being the signature that $\Re$ is the natural mass unit of the PDL framework [11]. Combined with the observation that $r_{\text{val}}/\Re = 155 \in \mathbb{Z}$ is a consequence of the unique solution $n_u = 24$ to the valence quadratic [9], the two lemmas force:

$$\delta\mu^{(1)} \;=\; \frac{r_{\text{val}}}{\Re \cdot R_{\text{tot}}} \;=\; \frac{155}{11017} \;\in\; \mathbb{Q}.$$

This derivation is entirely internal to PDL, invoking no QCD input and no gravitational parameter. It constitutes step (a) of the proof of the ε_G conjecture formulated in D28, reducing the remaining programme to step (b): establishing $2\Delta m_{\text{iso}}/m_p$ as the natural QCD correction forced by the $\Delta n = 4$ asymmetry of the PDL architecture.

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1 Introduction

The PDL framework [3, 5] reconstructs physical constants from four axioms on finite signed graphs. The minimal admissible stationary closure is the complete graph K_4 on four vertices and six edges, identified with the electron prototype with relational budget $\Re = 6$. The proton is the minimal hierarchical composite of such closures, uniquely selected by the quintuplet [8]:

$$(n_u,\ n_d,\ r_{\text{val}},\ R_{\text{sea}},\ R_{\text{tot}}) \ = \ (24,\ 28,\ 930,\ 10087,\ 11017). \tag{1}$$

Paper D28 [10] identified $155/11017 \in \mathbb{Q}$ as the leading-order valence engagement amplitude governing the proton-to-electron mass ratio correction, and stated its derivation from the PDL axioms as open problem OP2. The present paper resolves OP2.

The structure is as follows. Section 2 recalls the relevant PDL architecture. Section 3 proves Lemma 1 (stability condition for mixed triangles). Section 4 proves Lemma 2 (uniqueness of r_{val}). Section 5 assembles the main theorem. Section 6 states the remaining open problem and its connection to the ε_G conjecture.

2 PDL architecture: relevant definitions

2.1 Axioms and the K_4 closure

The PDL axioms select finite signed graphs $G = (V, E, s)$ with $s : E \to \{+1, -1\}$ satisfying four requirements: minimal binary pulsation (a global sign-flip $s \mapsto -s$ forming an irreducible two-cycle), triangular coherence (every triangle satisfies $s_{ij}s_{jk}s_{ik} = +1$), minimal completeness (self-referential closure without external support), and logical optimisation (maximisation of a coherence-weighted functional Ω). The unique minimal solution is K_4 [3, 6], with $\Re = \binom{4}{2} = 6$ edges and coherence defect $\nu(K_4) = 0$.

2.2 Proton architecture and the valence–sea partition

The proton is a two-tier composite. The *valence sector* G_{val} consists of three local stationary closures (two up-type cores of multiplicity $n_u = 24$ and one down-type core of multiplicity

$n_d = 28$), with total relational content:

$$r_{\text{val}} \;=\; 2\binom{n_u}{2} + \binom{n_d}{2} \;=\; 2 \times 276 + 378 \;=\; 930. \tag{2}$$

The *sea sector* G_{sea} comprises $R_{\text{sea}} = 10087$ dynamical relations whose sign configuration fluctuates independently of the valence pulsation cycle. Paper D9 [2] establishes that G_{sea} is precisely the set of relations for which pulsation-induced transient coherence violations are unavoidable: their cycle-averaged violation fraction $\Gamma_{\text{cycle}} > 0$ by construction.

2.3 Mixed triangles and the coupling geometry

The coupling between the proton and an electron-type K_4 is mediated by *mixed triangles*: coherent triangles (e_i, e_j, p_k) containing two electronic vertices $e_i, e_j \in V(K_4)$ and one proton vertex p_k. The sign of such a triangle is:

$$\sigma(e_i, e_j, p_k) \;=\; s_{K_4}(e_i, e_j) \cdot s_\times(e_i, p_k) \cdot s_\times(e_j, p_k), \tag{3}$$

where $s_{K_4}(e_i, e_j)$ is the sign of the edge (e_i, e_j) within K_4, and $s_\times(e, p)$ denotes the sign of a cross-edge linking an electronic vertex to a proton vertex. We define the *cross-product* at cycle c as:

$$P_c \;=\; s_\times(e_i, p_k, c) \cdot s_\times(e_j, p_k, c). \tag{4}$$

3 Lemma 1: stability condition for mixed triangles

Definition 3.1 (Stable mixed triangle)**.** A mixed triangle (e_i, e_j, p_k) is *stable* if it is coherent at both half-cycles of the K_4 pulsation:

$$s^{(c)}_{K_4}(e_i, e_j) \cdot P_c \;=\; +1 \quad \text{for } c \in \{1, 2\},$$

where $s^{(1)}_{K_4}$ and $s^{(2)}_{K_4} = -s^{(1)}_{K_4}$ are the two sign configurations of the K_4 pulsation.

Lemma 3.2 (Stability condition)**.** *A mixed triangle (e_i, e_j, p_k) is stable if and only if the cross-product satisfies:*

$$\text{(A)} \quad P_1 \;=\; s^{(1)}_{K_4}(e_i, e_j), \tag{5}$$

$$\text{(B)} \quad P_2 \;=\; -P_1. \tag{6}$$

Proof. The stability conditions at the two half-cycles are:

$$[C_1] \quad s^{(1)}_{K_4}(e_i, e_j) \cdot P_1 = +1,$$

$$[C_2] \quad s^{(2)}_{K_4}(e_i, e_j) \cdot P_2 = +1.$$

Since $s^{(1)}_{K_4} \in \{+1, -1\}$, equation $[C_1]$ gives immediately $P_1 = 1/s^{(1)}_{K_4} = s^{(1)}_{K_4}$, which is condition (A). Since $s^{(2)}_{K_4} = -s^{(1)}_{K_4}$, equation $[C_2]$ gives $P_2 = 1/s^{(2)}_{K_4} = -s^{(1)}_{K_4} = -P_1$, which is condition (B). Both conditions are necessary and sufficient by direct substitution. □

Remark 3.3 (Exhaustive verification)**.** Lemma 3.2 was verified exhaustively over all 8 coherent sign configurations of K_4, all 6 edges $(e_i, e_j) \in E(K_4)$, and all $2^4 = 16$ combinations of cross-signs at the two half-cycles. The 768 resulting cases yield zero counter-examples: the two counters for "(A∧B) and not stable" and "not (A∧B) and stable" are both identically zero. The equivalence is therefore not merely algebraic but structurally robust across all admissible pulsation configurations.

Remark 3.4 (Structure of the stability condition)**.** Conditions (A) and (B) depend only on the cross-product $P_c = s_\times(e_i, p_k, c) \cdot s_\times(e_j, p_k, c)$ and not on the individual cross-edge signs. There is therefore a single effective degree of freedom per half-cycle governing stability. Condition (A) is an *initial alignment* requirement: the cross-product must agree with the K_4 sign at cycle 1. Condition (B) is a *phase-locking* requirement: the cross-product must reverse exactly when K_4 reverses.

3.1 Consequence for the valence–sea partition

Proposition 3.5 ($\mathrm{G_{val}}$ satisfies (A)∧(B) structurally)**.** *Every relation of G_{val} that participates in a coupling with K_4 satisfies conditions* (A) *and* (B) *of Lemma 3.2 by construction.*

Proof. Condition (A): The active surface $R_{\mathrm{surf}} = 310\varphi$ is defined in the PDL corpus [3, 4] as the minimal envelope of relations forming *coherent* mixed triangles with K_4. Coherence at cycle 1 is the definition of membership in the active surface, which directly gives $P_1 = s^{(1)}_{K_4}$. Condition (B): The valence cores are local stationary closures admitting a binary pulsation of period 2 [3]. The PDL axiom of minimal binary pulsation requires this period to be minimal, hence equal to 2. The pulsation of each valence core is therefore synchronised with that of K_4, giving $P_2 = -P_1$. □

Proposition 3.6 ($\mathrm{G_{sea}}$ is excluded in the stationary regime)**.** *The contribution of G_{sea} to the stationary coupling amplitude with K_4 is zero.*

Proof. By definition [2], G_{sea} consists of relations for which $\Gamma_{\mathrm{cycle}} > 0$: their sign configuration has no fixed phase relationship with any stationary pulsation. A relation in G_{sea} satisfies conditions (A) and (B) simultaneously with probability 1/4 at any single cycle, as verified by exhaustive enumeration over the 16 cross-sign configurations. For a stationary regime extending over N independent pulsation cycles, the probability of maintaining (A)∧(B) is $(1/4)^N \to 0$ as $N \to \infty$. The time-averaged stable coupling amplitude from G_{sea} therefore vanishes in the stationary limit, leaving G_{val} as the sole contributor. □

4 Lemma 2: uniqueness of the valence engagement

Lemma 4.1 (Uniqueness of r_{val})**.** *Among all integer engagements of the form $a\,r_u + b\,r_d$ with $a \in \{0, 1, 2\}$ and $b \in \{0, 1\}$ (complete core engagements), the unique combination satisfying simultaneously:*

- *net charge $+1$, and*
- *divisibility by $\Re = 6$,*

is $(a, b) = (2, 1)$, giving $r_{\mathrm{val}} = 2r_u + r_d = 930$.

Proof. The charge of a core engagement is $a \cdot (+2/3) + b \cdot (-1/3) = (2a - b)/3$. The net charge $+1$ condition requires $2a - b = 3$. The six cases $(a, b) \in \{0, 1, 2\} \times \{0, 1\}$ yield charges $0, -1/3, 2/3, 1/3, 4/3, 1$. Only $(a, b) = (2, 1)$ gives charge $+1$. The engagement is then $2r_u + r_d = 2 \times 276 + 378 = 930$, and $930/6 = 155 \in \mathbb{Z}$. All other engagements either violate the charge condition or are zero. Exhaustive verification confirms the uniqueness. □

Remark 4.2 (Divisibility and the mass unit)**.** The divisibility condition $r_{\mathrm{val}} \equiv 0 \pmod{\Re}$ is not an independent assumption. Paper D27 [11] establishes that $\Re = 6$ is the natural mass unit of the PDL framework: the neutron mass gap $\Gamma_n = \Re\,\mu_n - R_{\mathrm{tot}}(n) = 40.102$ shows that PDL mass ratios are measured in multiples of $1/\Re$. The leading-order mass correction $\delta\mu^{(1)}$ must therefore be of the form $k/\Re$ for some integer k, which forces divisibility. That $930 = 6 \times 155$ is the unique integer satisfying both charge and divisibility is a consequence of the solution $n_u = 24$ to the valence quadratic established in D24 [9].

Remark 4.3 (Integrality of $r_{\text{val}}/\Re$)**.** The value $r_{\text{val}}/\Re = 930/6 = 155 \in \mathbb{Z}$ follows from the unique solution of the quadratic constraint derived in D24. Substituting $n_d = n_u + 4$ into $r_{\text{val}} = n_u(n_u - 1) + n_d(n_d - 1)/2$ and imposing the quasi-completeness condition yields $3n_u^2 + 5n_u - 1848 = 0$, with discriminant $22201 = 149^2$, giving the unique positive integer solution $n_u = 24$. One verifies directly that $930 = 6 \times 155$, confirming that the divisibility is a structural consequence of $n_u = 24$ and not an independent postulate.

5 Main theorem: axiomatic derivation of 155/11017

Theorem 5.1 (Leading valence engagement amplitude)**.** *The leading-order correction to the bare* PDL *mass ratio* $\mu_{\text{PDL}} = R_{\text{tot}}/\Re$ *is:*

$$\delta\mu^{(1)} = \frac{r_{\text{val}}}{\Re \cdot R_{\text{tot}}} = \frac{155}{11017} \in \mathbb{Q}. \tag{7}$$

This amplitude is the unique first-order valence engagement amplitude compatible with the PDL *axioms.*

Proof. The mass ratio correction $\delta\mu = \mu_{\text{PDL}} - \mu_{\text{exp}}$ arises from the coupling between the proton and K_4. By Proposition 3.5, only the relations of G_{val} contribute to the stationary coupling. By Proposition 3.6, the contribution of G_{sea} vanishes in the stationary limit. The available coupling budget per edge of K_4 is therefore $r_{\text{val}}/\Re$. Since $\mu_{\text{PDL}} = R_{\text{tot}}/\Re$, the leading-order fractional correction is normalised by R_{tot}, giving $\delta\mu^{(1)} = (r_{\text{val}}/\Re)/R_{\text{tot}} = r_{\text{val}}/(\Re \cdot R_{\text{tot}})$. By Lemma 4.1, $r_{\text{val}} = 930$ is the unique integer engagement satisfying net charge $+1$ and divisibility by $\Re$. By the remark following Lemma 4.1, $r_{\text{val}}/\Re = 155 \in \mathbb{Z}$. Hence $\delta\mu^{(1)} = 155/11017 \in \mathbb{Q}$. □

Table 1: Numerical verification of the main theorem against D28.

Quantity	Value	Source
r_{val}	930	Equation (2)
$\Re$	6	K_4 edge count
R_{tot}	11017	Quintuplet (1)
$r_{\text{val}}/\Re$	$155 \in \mathbb{Z}$	Lemma 4.1
$\delta\mu^{(1)} = 155/11017$	0.01406917	Theorem 5.1
$\delta\mu^{(1)}$ (D28 numerical)	0.01406900	[10]
Residual	11.8 ppm	—
$\delta\mu$ after QCD correction (D28)	0.013993	[10]
Residual after QCD correction	47 ppm	[10]

Corollary 5.2 (Resolution of OP2 in D28)**.** *Theorem 5.1 resolves open problem OP2 of paper D28 [10]: the amplitude* 155/11017 *is proved to be the exact first-order valence engagement amplitude from the* PDL *axioms applied to the proton–electron coupling geometry, without any empirical input.*

6 Connection to the ε_G conjecture

6.1 Recap of the D28 conjecture

Paper D28 [10] established the conjecture:

$$\varepsilon_G \stackrel{?}{=} \frac{\kappa \cdot \mathcal{C}}{6 + \kappa} \cdot \left(1 + \frac{2}{R_{\text{tot}}}\right), \tag{8}$$

accurate to 17 ppm against CODATA 2022, where $\kappa = R_{\text{surf}}/R_{\text{tot}} \in \mathbb{Q}(\sqrt{5})$ and $\mathcal{C} = (n_u^2 - 1)/n_u^2 = 575/576$. It further established that this conjecture holds if and only if $\mu_{\text{exp}} = \mu^* = 1836.152670$, and that $\delta\mu = (155/11017) \cdot (1 - 2\Delta m_{\text{iso}}/m_p) + O(47 \text{ ppm})$.

6.2 Two-step proof strategy

The proof of conjecture (8) reduces to two independent steps:

Step (a) — Derivation of 155/11017 **from** PDL **axioms.** This is resolved by Theorem 5.1 of the present paper.

Step (b) — Establishing $2\Delta m_{\text{iso}}/m_p$ as the natural QCD correction. This requires showing that the $\Delta n = n_d - n_u = 4$ asymmetry of the PDL proton architecture forces exactly the factor $2\Delta m_{\text{iso}}/m_p$ as the leading QCD correction to $\delta\mu^{(1)}$. The factor 2 corresponds to the number of up-type valence quarks in the proton (uud), and $\Delta n = 4$ is the same structural integer that governs the neutron mass gap $\Gamma_n = 40.102$ and the nuclear stability gap $(\Delta n+1)^2 = 25$ [7, 11]. Formalising this connection from the PDL axioms is the subject of the next paper in the programme.

6.3 Open problems

OP1 (Proof of the ε_G conjecture). Step (a) is now resolved. The remaining task is step (b): establishing $2\Delta m_{\text{iso}}/m_p$ as the structural QCD correction forced by $\Delta n = 4$.

OP2 (The 47 ppm residual). Paper D28 identifies the residual as consistent with QED corrections $\alpha \cdot \Delta m_{\text{iso}}/m_p \approx 50$ ppm. A structural derivation of this term from the PDL architecture would complete the derivation of $\delta\mu$ to full experimental precision.

OP3 (Stationary limit of $\mathbf{G_{sea}}$). Proposition 3.6 excludes G_{sea} via a probabilistic argument ($(1/4)^N \to 0$). A fully algebraic exclusion would follow from showing that the PDL axiom of minimal binary pulsation *defines* G_{sea} as the complement of the set of relations satisfying (A)∧(B) with respect to K_4, which is the definition of the active surface in the PDL corpus [3].

OP4 (Test via μ precision). The full D28 conjecture predicts $\mu = \mu^* = 1836.152670$. Any μ measurement below 0.001 ppm tests this independently of G [1].

Conclusion

This paper has proved that the leading-order valence engagement amplitude 155/11017 is the unique first-order correction to the PDL bare mass ratio compatible with the axioms of the framework. The proof rests on two algebraic lemmas. Lemma 1 characterises stable mixed triangles between K_4 and proton vertices by two conditions (initial alignment and phase-locking), verified exhaustively over 768 cases with zero counter-example. Lemma 2 identifies $r_{\text{val}} = 930$ as the unique integer valence engagement satisfying net charge +1 and divisibility by $\Re = 6$, the latter being the natural PDL mass unit established in D27. Together they force $\delta\mu^{(1)} = r_{\text{val}}/(\Re \cdot R_{\text{tot}}) = 155/11017 \in \mathbb{Q}$, resolving open problem OP2 of paper D28.

The derivation is entirely internal to PDL: it invokes no QCD mass input, no gravitational parameter, and no cosmological observation. It constitutes step (a) of the two-step proof of the ε_G conjecture, and reduces the remaining programme to step (b): formalising the connection between the $\Delta n = 4$ structural asymmetry and the QCD isospin correction $2\Delta m_{\text{iso}}/m_p$.

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